12/9/60

The proposed Simulation and Computing Center would serve the following three purposes:

* Enhance large-scale systems development through simulation at various levels throughout the development cycle.
* Support CCDD Systems - Oriented Applied Research -
* Provide ultra-large scale computing facilities for the Boston-area Air Force research community

These needs are detailed separately below:

SYSTEMS SIMULATION

The digital computer has become recognized in the last decade as a powerful and extremely valuable tool in the development of large-scale systems. The cost of digital simulation has been more than justified by the staggering costs of ineffeciencies in the vast complex of men and equipment that is the end product of the development effort . It is also justified by the cost in time and money of abortive development efforts that could have been terminated or redirected earlier had the system problem been better understood.

CCDD has been charged with the responsibility for directing the development of not just one but approximately one dozen of most complex large-scale systems in the world. This document proposes a simulation facility commensurate with that responsibility.

The uses of a digital computer simulation facility in large-scale system development are many and often interconnected. Some of these uses are presented below ordered roughly according to the phase of the development cycle.

WAR GAMING

The effectiveness of many postulated Air Force systems is difficult to evaluate since, quite fortunately, the time operational environment has never existed . Because of this the environment has often been simulated on a digital computer using war gaming techniques. General simulation of global and limited war situations is a valuable tool for the evaluation of pertinent large-scale systems. With appropriate computer program design, the influence of projected system parameters upon the course of the battle can be traced This enables a realistic measurement of the effectiveness of the projected system. Computer-simulated environments are also valuable in evaluating other systems where the costs of coupling into the actual environment are prohibitive.

With the availability of sufficiently-flexible environmental models in computer-program form postulated systems can be tested very soon after their conception. A rough determination of the value of the system can thus be made.

As systems design proceeds more specific and pertinent environmental models may be developed. Simulations using these models will enable a close estimation of the payoffs due to changes in system parameters.

GENERALIZED SYSTEM SIMULATORS

General-purpose system simulation programs are a relatively development. Several such programs have been recently developed within IBM in order to study complex queueing situations.

These queueing problems cannot presently be solved analytically and represent a major problem area in systems design. One such IBM-developed general-purposed simulator has contributed to the solution of such diverse problems as optimizing the flow of customers through a supermarket and to computer configuration design for automated airline reservation systems.

General-purpose simulator programs give the system designer a powerful tool which is immediately available at the inception of system design.

SYSTEM SIMULATION

As the system design takes shape and equipment is initially specified more detailed and specific simulation can take place. The intricate nature of the man-machine complex being designed necessitates detailed simulation of the manifold information flows and processes if the highest performance at the lowest cost is to be expeditiously achieved.

The initial system simulation program is usually relatively simple and hopefully quite flexible. The main interest is first order effects of variations in system configuration and major parameters. More detailed models are developed later in the development cycle. The systems simulator program is written so that it may be coupled to other simulator programs such as the environmental model discussed earlier and to real-life inputs from operators and sensors as such coupling becomes appropriate.

SUBSYSTEM SIMULATION

In addition to overall systems models used to directly evaluage and design the system as a whole, subsystem models will also be useful in the design and evaluation of parts of the system. Specific subsystem problems can be isolated and resolved through the use of digital simulation.

REAL-TIME SIMULATION

As system development continues it becomes possible and desireable to couple actual pieces of the ultimate system into the digital simulation activity. These "pieces" include both equipment and operators. This simulation places two burdens upon the computing equipment:

(1) The time-scale of the simulation is constrained to real-time, the natural time-scale of the operators and system components. This means that extremely high speed computation may frequently be require . In other situations the computational bur den may be extremely low and it is desireable to have the capability to interleave this activity with other computational jobs.

(2) A large and flexible input/output system is required to meet various simulation requirements.

SYSTEM TESTING

The build-up of simulation programs and facilities during the development cycle will reach the point where the system is being tested and evaluated as a detailed whole. This will frequently require tremendous data processing capacity. The real-time constraint

will be present. Many of the pieces of the system may be coupled through the input/output facilities of the simualtion complex. Much of the systems environment and frequently parts of the system must be simulated digitally. Finally, results must be recorded for later evaluation.

SYSTEM IMPROVEMENT

The evolutionary nature of Air Force Command and control systems usually requires the continuing existance of a test and simulation facility for the development of system improvements. Thus, the requirement for system simulation exists for the operational life of the system.

CCDD APPLIED RESEARCH

The Technology Directorate of CCDD has the responsibility of managing and performing applied research directly associated with command and control systems. The availability of a large and flexible computer complex will do much to further this effort. Such areas man-machine communication, decision-making both human and mechanical, and computer programming for command and control systems are quite critical. A flexible large-scale computer complex is an invaluable too in these areas.

HANSCOMB COMPLEX COMPUTING CENTER

The Hanscomb Complex (CCDD, CCDD50, 3245th Air Base Wing, CRC, Lincoln Laboratory, The MITRE Corporation, etc.) includes seve al large computer users. It is anticipated that the needs of these organizations from automatic scientific computation

will continue to grow at a rapid rate. The existance of an ultra-large-scale computer complex at CCDD would enable the computational needs of these important Air Force-oriented research and development organizations to be met in an efficient and economical manner.

It has long been recognized that large computers are more economical on a cost per computation basis than small computers. The Hanscomb Complex will have sufficient work to keep a very large computer complex busy and thereby reap this economic reward.

It is also well recognized that large-scale computers make the programming job much less difficult. The small machine constraints of small data storages, limited input/output, and limited instruction sets are greatly lessened. The programmer can then use less of his energy and creative ability in getting-around machine constraints and more towards getting the job done simply and effeciently.

Many of the research and development tasks being pursued in Hanscomb Complex organizations make an extremely fast and flexible computer quite desireable. Euristic research, in particular, is an area where there seems to be no practical upper bounds on desired computing speed and data storage capability. In almost all areas of scientific calculation, the availability of greater computing capability allows a finer and more detailed analysis and therefore improved results.

Proposed Computer Complex

In order to meet the large and rapidly growing computational needs of the Hanscomb Complex, IBM is pleased to propose a data processing system of unprecedented power, flexibility and growth potential. So that CCDD may rapidly arrive at a satisfactory initial configuration, three options are proposed. These are outlined below:

Option A

Option A consists of an IBM STRETCH Data Processing System. Developed for the AEC's Los Alamos Laboratory, to be the most powerful system available today. In addition to Los Alamos, the following highly-experienced computer users have contracted for STRETCH Systems:

Corporation for Economic and Industrial Research (CEIR)

United Kingdom Atomic Energy Commission

The STRETCH configuration proposed under Option A includes the following elements:

IBM --- Central Processing Unit

- Fastest available computing speeds
- An instruction set of unprecedented flexibility

IBM --- Core Storage Unit (- Units)

- 2.2 microsecond cycle
- 16,384 64-bit words per unit
- automatic error detection and correction

IBM

Basic Exchange

Handles up to _____ computer input/output operations simultaneously

Standard interface Channels enable attachment of a wide range to digital data acceptors or donors

Option B

Option B consists of the STRETCH system outlined in Option A above plus three Satellite Computer systems. The Satellite Computer can be described as a small version of STRETCH, offering many of the advanced design features and flexibility of STRETCH at a lower cost. It also offers many compatibility features having the same memory word sizes, basic character size, input/output interface, and a very similiar design philosophy. In Option B, the three Satellite Computers are connected to STRETCH via the standard input/output facilities of each machine. Digital information (programs or data) may be transmitted rapidly (bits per second) and automatically under stored program control between each Satellite and STRETCH. Option B augments the Option A list of elements with the following:

Option C

Option C consists of the same basic elements as Option B, the STRETCH System and three Satellite Computer Systems. Option C differs from Option B in that the Satellite Computers sharg Core Storage Units of the STRETCH System. This does away with the standard input/output channels and greatly increases the flexibility and speed of inter-computer communication. The fact that either a satellite or STRETCH or both (on a time-sharing basis) can use certain core storage units greatly increases the flexibility and usefullness of the computer complex.